Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
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Momentum Unleashed:

Predicting the "Snowball Effect" in Competitive Sports

Abstract

Nowadays, with the increasingly fierce competition in matches, the outcomes are increasingly uncertain. However, it is certain that the incredible swings usually attributed to "momentum" are becoming more and more common, just as what has happened in the 2023 Wimbledon Gentlemen's final. So, it is of significance to strengthen the understanding of momentum and its corresponding impact on the matches. In this paper, we aim to establish a model from a more comprehensive perspective to help predict momentum changes and offer reasonable suggestions applicable to future training.

Firstly, we review relevant literature and form a three-category system of **13 indicators** related to the players' performance. Then after the pre-processing and normalization of data provided, we carry out the **Principal Component Analysis (PCA)** to achieve the dimensionality reduction and **Analytic Hierarchy Process (AHP)** to determine the weights of the principal components. Next, we establish the **Players' Performance Evaluation Model (PPEM)**. Finally, we conduct data visualization based on PPEM, which successfully depicts the changes and comparisons in players' performance.

Secondly, we assess the randomization of momentum impact from two aspects i.e., the swings in play and runs of success. For the swings in play, we apply **Monte-Carlo Simulation** to get the distribution of simulate scores. Then we conduct **Kolmogorov-Smirnov test (KS Test)** and identify the differences in the distributions of the samples of real scores and simulate ones. For the runs of success, we analyze the relationship between players' performance and the point difference to conclude that consecutive points and momentum have mutual influence.

Thirdly, we categorize momentum into **Strategic Momentum (SM)** and **Psychological Momentum (PM)**. For SM, we creatively combine the **Win-loss index** and **Ranking** based on the theory of SM. For PM, we find it is highly related to the players' performance through literature study. Consequently, we apply **Categorical Boosting (CatBoost)** to optimize the PPEM by selecting **Server, Winning Streak, Speed** and **other 4 significant indicators** and adjusting the indicators' weights to establish the **Adjusted PPEM** to depict PM. Based on the combination of SM and PM, we successfully establish the **Momentum Prediction Model (MPM).** Finally, based on MPM, we help coaches provide players with suggestions to solve the "3N" problems (New opponent, New match, New situation) to deal with the differential momentum swings in the past matches.

In the end, we apply data collected from one match in tennis of **women's single on clay court** and one **Olympics match in table tennis** of men's single and verify the great generalization ability of the MPM. For a few of poor predictions, we propose some ways to optimize in the future model.

Keywords: Momentum; PCA; AHP; Monte-Carlo Simulation; CatBoost

CONTENTS

1 Introduction

"The beauty of competitive sports is uncertainty."

—Harimoto Tomokazu, Japanese Table Tennis Champion

1.1 Problem Background

While watching a match on the spot or through a TV broadcast, we are accustomed to forming our own judgement on the field situation based on the match analysis with various information provided or on the screen. Sometimes it is not strange for audience to witness the incredible swings for many points or even games. Just as what has happened during the 2023 Wimbledon Gentlemen's final, Novak Djokovic, was defeated by a young up-and -comer Carlos Alcaraz, even though it is Djokovic who had a more competitive edge at the beginning set.

Analogy is usually made when this kind of uncertainty occurs. In Physics, the strength or force gained by motion or by a series of events is defined as "momentum". This similar implication can be applied to sports. The strength during a match felt by or a player that help him take advantage can also be attributed to "momentum". Consequently, understanding, calculating, and predicting player momentum is of significance to figure out changes and trend during a match or game. By conducting in-depth analysis of every point from all Wimbledon 2023 men's matches after the first 2 rounds, we will comprehend how various events during the match act to create or change momentum, which can also be helpful in future player-coaching.

1.2 Restatement of the Problem

To provide advice for coaches on what is the role "momentum" playing and how to get players prepared to event that impact flow of play during matches, we need to solve following problems by model construction and application of momentum according to the given data:

(1) Propose a method to measure the momentum during a match or game.

(2) Figure out how various events during the match act to create or change momentum.

To address the above two problems, we need to solve the specific tasks as follow:

- **Task 1:** Establish a model to describe changes in flow of play when points occur and figure out which player is performing better and how much better he is performing.
- **Task 2:** Assess the claim that the role momentum's play on the player's success is random, instead of related.
- **Task 3:** Conduct in-depth analysis to figure out indicators that can be employed to predict change of the play flow in favoring one player to the other
  - a) Develop a model to make predictions of the swings in the match and figure out what factors seem most related.
  - b) Give suggestions to player going into a new match against a different player.
- **Task 4:** Test the model on other differential matches and optimize it by taking more indicators into consideration to formulate a more generalizable one.

1.3 Literature Review

The relationship between momentum and performance has elicited the curiosity of athletes, coaches, and sport psychologists since the late 1970s [1]. The scholars have utilized different methods to carry out analysis of data collected from different matches to shed light on the effects momentum have on the outcome of matches.

For the relationship between momentum and the players' performance. Stanimirovic and Hanrahan found that players, coaches, and spectators all consider psychological momentum as a determinant of success (2004) [2]. Recently, Helmut Dietl and Cornel Nesseler (2017) [3] found players benefit from momentum if they control a match. Ben Moss and Peter O'Donoghue (2015) [4] pointed out that momentum in tennis with breaks of serve in games have a significant influence on the outcome of the next game. Walid Briki (2017) [1] formed a more integrative perspective and empirically tested that psycho-behavioral momentum (PBM) can mediate the relationship between early and subsequent success.

For the different methods carried out to figure out the relationship. In 2009, Niksa Djurovic et al [5] standardized the variables and conducted a factor analysis under a component model to determine significant factors. In 2015, Ben and Peter used statistical analyses, including chi-square test and ANOVA to arrive at their conclusion. In 2017, Helmut and Cornel [3] utilized a binary approach and a set score approach in their analysis. In 2019, Anna Fitzpatric [6] developed a method, named the Percentage of matches in which Winner Outscored the Loser (PWOL), to test validity against traditional statistical methods like paired t-tests.

However, there is still a lack of a more precise quantitative model to help players, coaches and even audience to measure the momentum. **Furthermore**, there is also a time lag in the conclusions concerning the relationship momentum and performance. **Consequently,** the marginal contribution of our model is to establish a model, comprehensively considering momentum and related indicators, to assist in predicting the swings and changes in the match.

1.4 Our Work

In this paper, we develop two mathematical models progressively and address 4 problems step by step. More specifically, the flow chart of our work is presented as follows:

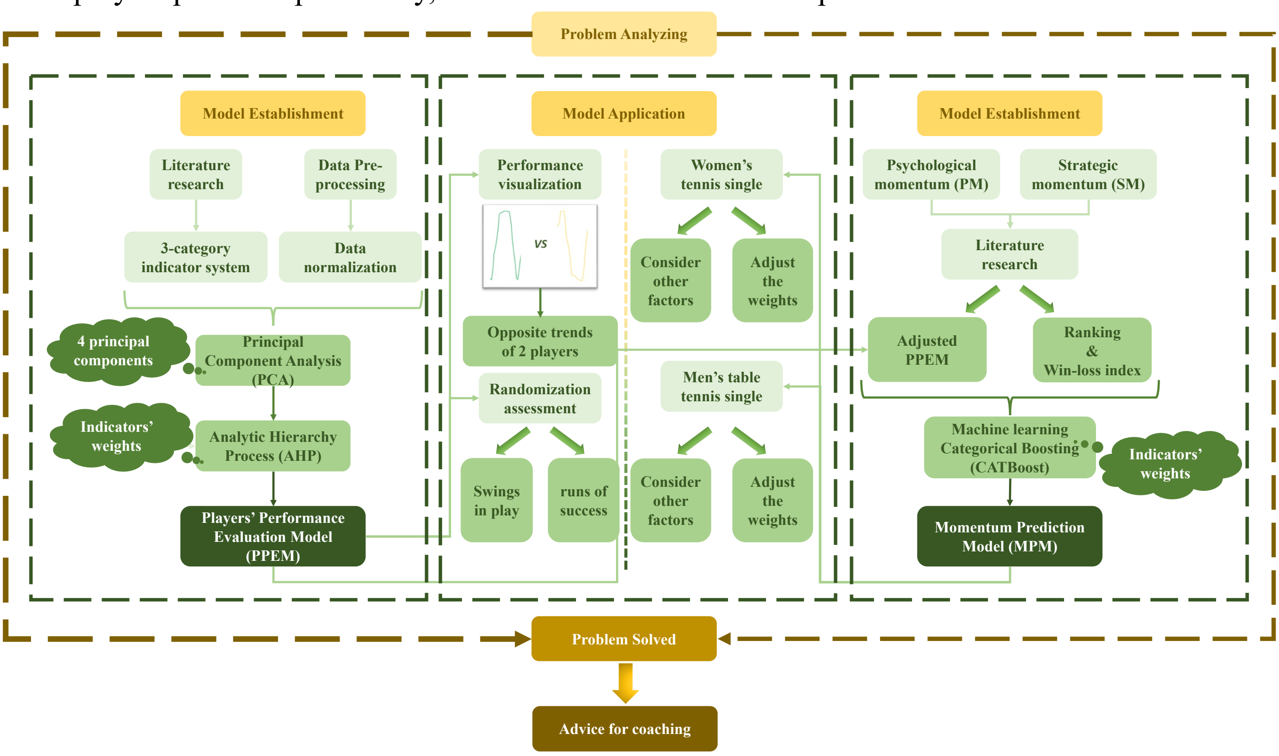

Figure 1: The Flow Chart of Our Work

2 Assumptions and Justifications

To simplify our problems, we propose the following basic assumptions, each of which is properly justified as below.

- **Assumption 1: All data sources collected and used in this paper are reliable.**
  Justification: We need to rely on historical, data such as the ranking, of each player to evaluate their momentum. Therefore, the reliability of data is very important.
- **Assumption 2: Players' performance and momentum can be comprehensively and accurately reflected through the indicators carefully selected.**
  Justification: Although the players' performance and momentum are determined by many factors, some of them are eliminated by PCA or other methods without much influence on the evaluation. Then an indicator system is formed that can fully and accurately present the evaluation.
- **Assumption 3: In addition to the data we can collect, performance and momentum are not affected by other factors, such as temporary injuries, home and away, and crowd feedback, etc.**
  Justification: In the establishment of the models, we should take the indicators that is common or can be easily measured. Some other rare or fortuitous indicators should be eliminated from the models.
- **Assumption 4: The player's strategic behavior (e.g., serving Angle, hitting forehand, etc.) is independent of momentum and performance.**
  Justification: As a professional athlete, the way to receive a serve that requires a long time to think about should be more beneficial to yourself, rather than being carried away by the momentum hidden in defeat or victory.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Key Notations Used in Paper

Symbol	Description
F_{count}	The scores of principal components
$W_{player\ num}$	The value vectors of players in PPEM
$PP_{player\ num}$	The final performance score of the player after moving average for each point
x_{ij}	The data after pre-processing and normalization
w	The weights of the indicators'
$Score_{player\ num}$	The initial performance score of the player
PM	The psychological momentum
SM	The strategic momentum

4 Players' Performance Evaluation Model

The evaluation of players' performance is to capture the flow of the play at the occurrence of points. It can output a quantitative result representing one player's general performance. If we consider one player edges over the other player in a match, the quantitative result of the former is always bigger than that of the later.

The **Players' Performance Evaluation Model (PPEM)** we are going to establish ought to meet the following requirements:

- The model can identify which player is performing better and how much better that the other quantitatively.
- The model should be comprehensive, containing various indicators related to the scoring of the players and indirectly related to the players' performance.
- The model is supposed to be robust, and the evaluation results are supposed to be relatively stable with the possible disturbance of uncertainties.

4.1 The PPEM Indicator System

4.1.1 Determination of the Indicators

To build the PPEM indicator system, we need to select representative and comprehensive indicators. In our previous literature reading and further research, we summarize different indicators of players' performance. In the process of statistical analysis, we find that some indicators have more outliers than others. Through comparison and categorization, we select 13 indicators in terms of reasonability to form the framework of our indicator system. **However,** for the indicators chosen, we tend not to address the outliers. Because the outliers here are more related to mathematical field, they are not the case when it comes to the real match field.

Our indicator framework is presented as follows:

Fatigue Level: The phenomenon of temporary physiological function decline caused by a large amount of exercise during the match. There are 2 indicators related to this category.

Personal Technic: The ability of an athlete to use technique in a particular sport, including correctness, stability, and proficiency of technical movements. The lead during the match usually reflects the player's personal technic. There are 5 indicators related to this category.

Real-time Mentality: The immediate mental state and emotional response of the athletes during the competition, including level of confidence, anxiety, concentration, etc. There are 6 indicators related to this category.

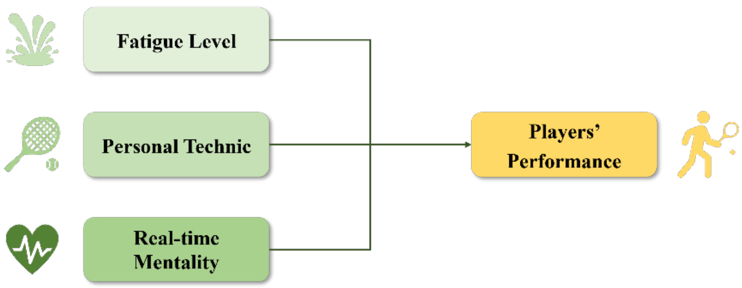

Figure 2: The PPEM Indicator Framework

Based on this framework, our PPEM indicator system is presented as follows:

Table 2: The PPEM Indicator System

Category	Indicator	Explanation	Type
Fatigue Level	Speed	Service speed	P
	Running Dis	Distance run in the last 3 minutes	N
Personal Technic	Leading Sets	Difference in winning sets	P
	Winning Streak	Winning streak points	P
	Point Dif	Scoring gap	P
	Unf Err	Number of unforced errors in the last 3 points	N
	Leading Game	Difference in winning games	P
Real-time Mentality	Deciding Set	Whether it is a deciding set or not	P
	Server	The receiving/serving side	P
	Double Fault	Number of double errors in the last 3 points	N
	Break Pt	Whether it is at break point or not	P
	Break Pt Missed	Whether to miss a break point chance or not	N
	Deciding Game	Whether it's a tiebreaker or not	P
Note: P: Positive Attributes; N: Negative Attributes			

4.1.2 Data Pre-processing

Data Filling: After a preliminary analysis of the data provided, we discover that only “Speed” has some missing values. To solve this problem, we adopt the following measure.

- For the few missing data, we find the indicator has a relatively strong correlation with the year indicator. As a result, we use the regression interpolation method to complete data filling.

Addressing Outliers: Through descriptive statistics, we analyze each indicator in the match and find some abnormal outliers that deviated from the mean by more than twice the standard deviation.

- For the outliers, we find there is only a small fraction of outliers compared with the large quantities of data provided. Hence, these outliers can be neglected.

4.1.3 Data Normalization

Now that we have got a comprehensive and accurate dataset, we need to normalize the data of different indicators so that they can be compared on the same scale. The 13 indicators can be divided into two categories, and we conduct different methods for normalization accordingly. For most of the indicators, it is quite clear whether they are positive attributes or negative attributes.

- Positive Attributes: the larger, the better.

$$\tilde{x}_{ij} = \frac{x_{ij} - min\{x_i\}}{max\{x_i\} - min\{x_i\}} \tag{1}$$

- Negative Attributes: the smaller, the better.

$$\tilde{x}_{ij} = \frac{max\{x_i\} - x_{ij}}{max\{x_i\} - min\{x_i\}} \tag{2}$$

After the data normalization, firstly, we compare the 2 players performance in different indicators and visualize the specific comparison on the figures as follow:

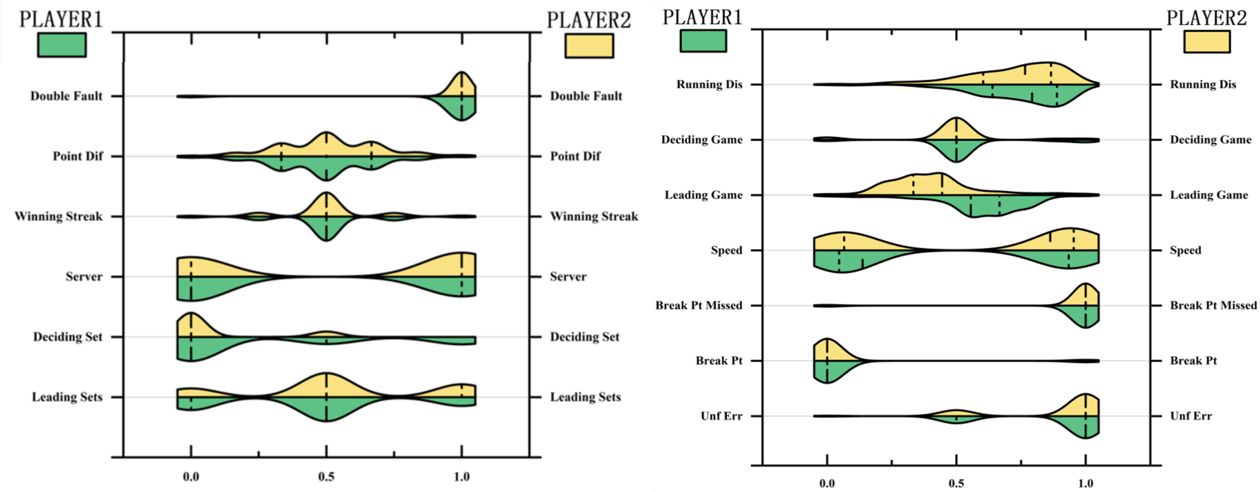

Figure 3: The Players' Performance with Different Indicators

From these 2 figures, we can vividly see that some comparisons and technic gaps between the 2 players. For example, the "Speed" indicator, the speed of ball for player 1 in much bigger than that of player 2. Furthermore, the "Leading Game" indicator, we can easily find that player 1 has more leading games during the match.

After the visualization of the players' performance in different indicators, we analyze the inner correlation of our indicators in the PPEM indicator system. We draw a heat map of the correlation between the indicators and the darker the color is, the stronger the correlation the two indicators have.

Figure 4: The Heat Map of the Indicators' Correlation

From this figure, we can vividly see that some of the indicators have a strong correlation. For example, the "Speed" indicator and the "Serve" indicator. It means that the player who is on the serve is inclined to serve at a relatively higher speed, which is quite reasonable to real life because most player wants to score on his own serve. Furthermore, the "Unf Err" indicator and the "Double Fault" indicator also have a strong correlation. It is quite common in many matches that the player who have more double errors in the last 3 points is apt to have an unforced error due to the fluctuation of mentality during the game.

4.2 The Establishment of PPEM

We introduce the Players' Performance (PP) as the overall description of the play flow by utilizing the combination method of **Principal Component Analysis (PCA)** and **Analytic Hierarchy Process (AHP)**. The main procedure is presented as follows:

The First step, conduct PCA to achieve dimensionality reduction based on the new categorized indicator system. A few indicators with large variance contribution rate are derived from the original variable, so that they can retain the information of the original variable as much as possible. Consequently, simplify the third-level indicators.

The Second step, explore the correlation between variables through literature analysis and classify the original variables, i.e., the variables with high correlation are divided into a group, and then based on the results of the above principal components, the cross-influence of the third-level indicators on the second-level indicators is re-examined. Hence, a more scientific and reasonable multi-level target system is revised.

The Third step, construct scoring matrix objectively and accurately. Then, the weight of each index can be obtained according to the score matrix. After the index value of each index is dimensionless, the comprehensive score value of each sample can be obtained by linear comprehensive evaluation method, i.e., the evaluation score value of each index is multiplied by the corresponding weight.

4.2.1 Principal Component Analysis of Indicators

Since there may be internal correlation among the 13 indicators, we first conduct principal component analysis on these indicators to achieve the dimensionality reduction and eliminate the collinearity between variables. The analysis steps are presented as follows:

Assuming that there are n samples and p indices, a sample matrix X of $n \times p$ can be formed.

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \tag{3}$$

Step 1: Calculate the covariance matrix R of the matrix. The data normalization has been conducted before, so we can directly skip to the next step.

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & r_{pp} \end{bmatrix} \tag{4}$$

For r_{ij} in matrix R, we have:

$$r_{ij} = \frac{1}{n-1}\sum_{k=1}^{n}(x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j) = \frac{1}{n-1}\sum_{k=1}^{n} x_{ki}x_{kj} \tag{5}$$

$$R = \frac{\sum_{k=1}^{n}(x_{ki}-\bar{x}_i)(x_{kj}-\bar{x}_j)}{\sqrt{\sum_{k=1}^{n}(x_{ki}-\bar{x}_i)^2\sum_{k=1}^{n}(x_{kj}-\bar{x}_j)^2}} \tag{6}$$

Where $i,j = 1,2,\dots,p$

Step 2: Calculate the eigenvalues and normalized eigenvectors of the covariance matrix.

For eigenvalues λ_i of the covariance matrix, $\lambda_1 \gg \lambda_2 \gg \cdots \gg \lambda_p \gg 0$, matrix R is a positive semi-definite matrix and $tr(R) = \sum_{i=1}^{p}\lambda_i = p$.

For normalized eigenvectors A_i of the covariance matrix, we have:

$$A_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{p1} \end{bmatrix}, A_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{p2} \end{bmatrix}, \dots, A_p = \begin{bmatrix} a_{1p} \\ a_{2p} \\ \vdots \\ a_{pp} \end{bmatrix} \tag{7}$$

Step 3: Calculate principal component contribution rate and cumulative contribution rate.

$$\text{principal component contribution rate} = \frac{\lambda_i}{\sum_{k=1}^{p}\lambda_k}(i = 1,2,\dots,p) \tag{8}$$

$$\text{cumulative contribution rate} = \frac{\sum_{k=1}^{i}\lambda_k}{\sum_{k=1}^{p}\lambda_k}(i = 1,2,\dots,p) \tag{9}$$

Step 4: Derive all principal components from the computed eigenvalues and eigenvectors.

$$F_{count} = \sum_{k=1}^{p} a_{ki}x_{ki} \tag{10}$$

F_{count} are the principal components, a_{ki} is the (i,j) elements of the covariance matrix.

Step 5: Select the number of principal components by determining whether the cumulative contribution rate is greater than 80%. For a certain principal component, the greater the coefficient in front of the index, the greater the influence of the index on the principal component.

After conducting these steps, we get 4 principal components $F_{count}(count = 1,2,3,4)$.

4.2.2 Analytic Hierarchy Process of Indicators

According to the conclusion of PCA and combined with the PPEM indicator system, we analyze the relationship between the factors in the system and the establish hierarchy structure of the indicators.

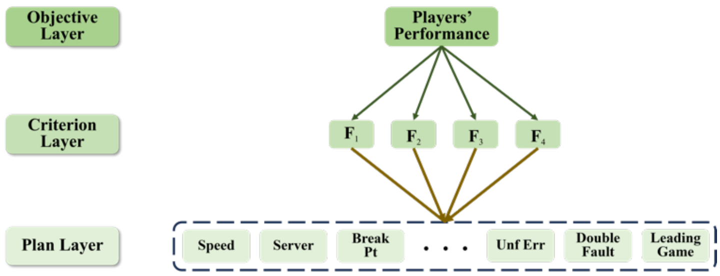

Figure 5: The Hierarchy Structure of Indicators

Step 1: Establish the three-level hierarchy structure of the indicators. The decision problem is divided into three levels. The uppermost layer is the objective layer representing the players' performance. The lowest layer is the plan layer, namely 13 indicators. The middle layer is the criterion layer including 4 principal components achieved by the PCA method. In order to have a clearer figure, we draw all of the 13 lines between F_1 and indicators in the third-level and set the 13 indicators as a whole when drawing the lines between the F_2 /F_3 /F_4 and the whole. The specific hierarchy structure is presented as follows:

Step 2: Conduct consistency test. We calculate 2 indicators here, namely the Consistency Index (CI) and the Consistency Ratio (CR).

For CI, we have:

$$CI = \frac{\lambda_{max} - count}{count - 1} \tag{11}$$

For CR, we have:

$$CR = \frac{CI}{RI} \tag{12}$$

Here, λ_{max} stands for the maximum eigenvalue of the judgment matrix and the RI stands for Random Index. We query the RI table to determine the RI value corresponding to a certain number of indicators. The RI table is presented as follows:

Table 3 The Value of Random Index

$count$	1	2	3	4	5	6	7	8	9	10	11	12	13
RI	0	0	0.52	0.89	1.12	1.36	1.41	1.46	1.49	1.52	1.54	1.56	1.58

As we have 13 indicators in our PPEM indicator system, the value of RI is 0.89. Then we substitute the specific data into the formula for calculation. We have:

$$CI = 2.57 \times 10^{-6}, CR = 2.88 \times 10^{-6} \tag{13}$$

Both the values of CI and CR are less than 0.1. Consequently, the calculation results have passed the conformance test.

Step 3: Construct judgment matrix. We conduct pairwise comparison of each element of the same level with respect to the importance of a criterion in the previous level.

For judgment matrix O-C: Compare the three elements $F_{count}(count = 1,2,3,4)$ of layer C in pairwise to obtain a comparison matrix. For each tennis match, there are 2 players. So, we analyze the matrix of the 2 players separately. The matrix of each player is presented as follows:

Table 4: The Judgement Matrix O-C for Players

Matrix O-C	F1	F2	F3	F4
F1	1.0000	2.3442	7.1266	8.0715
F2	0.4266	1.0000	3.0401	3.4432
F3	0.1403	0.3289	1.0000	1.1326
F4	0.1239	0.2904	0.8829	1.0000

Step 4: Calculate the weights. Based on the contribution values of different indicators and combined with the experts' rating, we calculate the weights w_{count} for $F_{count}(count = 1,2,3,4)$ and obtain a linear comprehensive evaluation method, i.e., PPEM. The model is presented as follows:

$$Players'Performance(PP) = \sum_{count=1}^{4} w_{count}F_{count} \tag{14}$$

4.3 The Application of PPEM

After the successful establishment of PPEM, we apply PAC and AHP to our data and get the specific weights of indicators. The table of different indicators' weights for players is presented as follows:

Table 5: The Different Indicators' Weights for Players

	F1	F2	F3	F4
Contribution Value	0.4886	0.2084	0.0686	0.0605
Leading Sets	0.0928	0.1798	-0.0011	0.0047
Deciding Set	-0.0054	0.1498	0.0025	0.0023
Server	0.3761	-0.0038	-0.0044	0.0019
Winning Streak	0.0821	0.1113	-0.0144	-0.0153
Point Dif	0.0522	-0.0045	-0.0098	-0.0020
Double Fault	-0.0172	0.0011	0.0092	0.0025
Unf Err	0.0623	-0.0057	0.1257	0.0383
Break Pt	0.1220	-0.0127	0.0246	-0.0074
Break Pt Missed	-0.0194	0.0021	0.0036	-0.0013
Speed	0.3041	-0.0033	0.0035	0.0018
Leading Game	0.0838	0.0250	-0.0319	0.0686
Deciding Game	-0.0145	0.0070	0.0115	0.0314
Running Dis	-0.0013	0.0018	-0.0022	0.0116

From the above table, we get the weight value vectors as follow:

$$W_{players} = (0.5915,0.2523,0.0828,0.0734) \tag{15}$$

Consequently, we calculate the overall evaluation of players' performance and establish the **Players' Performance Evaluation Model (PPEM)** as follows:

$$Initial\ PP_{players} = \left(F_1,F_2,F_3,F_4\right)W_{players}^T = (Score_1, Score_2, \dots, Score_p) \tag{16}$$

Furthermore, considering that a player's current performance is influenced by their performance in recent innings, and in order to reduce the impact of these random fluctuations in a single game on the overall evaluation, we combined the Simple Moving Average (SMA) method to determine a player's final performance score, thereby revealing trends in player performance.

Employ PPEM to calculate the time series data of $Initial\ PP_{player1}$ and $Initial\ PP_{player2}$. We set the window k value to 5 and get the final performance score of the player after moving average for each point x_t (where $t \geq 5$)

$$PP_{player1} = \frac{1}{5}\sum_{k=t-4}^{t} Score_{i\ player1} \tag{17}$$

$$PP_{player2} = \frac{1}{5}\sum_{k=t-4}^{t} Score_{i\ player2} \tag{18}$$

4.4 The Visualization of Players' Performance

To test whether our PPEM can truly reflect 2 players' performance and thus shows changes in players' scores, we visualize the performance of players in the first match of the 2023 Wimbledon Gentlemen's. The changes and comparisons in players' performance are shown above.

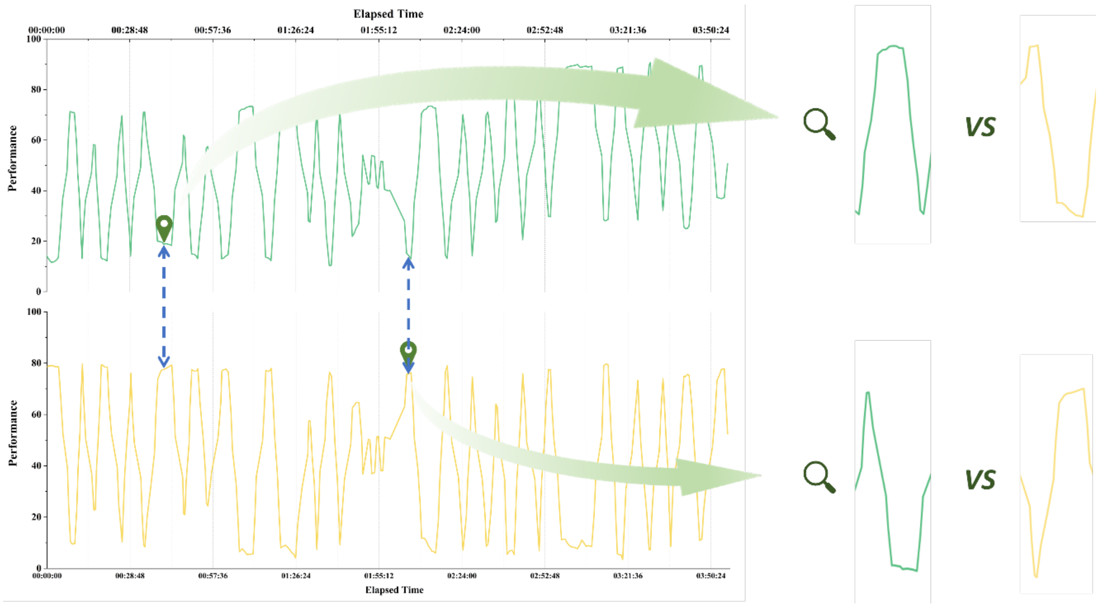

Figure 6: The Changes in Players' Performance Evaluation in the First Match

For the first comparison, we consult the provided dataset of each match's scores and have the changes of scores as follow (It is approximately 30 minutes after the beginning of the match):

Table 6: The Score Changes of the Players in the First Comparison

Players	37:18	37:44	38:26	38:54	39:29	40:01	40:33	41:23	42:31	43:18	43:59	44:29
Carlos Alcaraz	0	15	15	30	30	40	40	AD	40	40	40	AD
Nicolas Jarry	0	0	15	15	30	30	40	40	40	AD	40	40

From 37:18 to 38:54, we notice that Carlos Alcaraz wins consecutive points and the changes in his performance have a similar trend. Carlos's performance line increases while that of Nicolas decreases at the same time. Then, **from 39:29 to 43:18**, Nicolas wins consecutive points and gets an AD. The performance of him evaluated by the PPEM also has a rise. Finally, **from 43:59 to 44:29**, Carlos has an AD and thus performance of him has a rise while performance of Nicolas witnesses a drop.

For the second comparison, we consult the provided dataset of each match's scores and have the changes of scores as follow (It is approximately 120 minutes after the beginning of the match, so the denotation of time, such as "2:08:41", is presented as "08:41"):

Table 7: The Score Changes of the Players in the Second Comparison

Players	08:41	09:28	10:00	10:55	11:35	12:15	12:53	13:38	14:08	14:53	15:48	16:15
Carlos Alcaraz	0	15	15	15	30	40	40	AD	40	40	40	40
Nicolas Jarry	0	0	15	30	30	30	40	40	40	AD	40	AD

From 08:41 to 09:28, we notice that Carlos has a better beginning of this set, so his performance line generally has an upward trend. Then, **from 10:00 to 11:35**, Nicolas wins 30 points faster than Carlos, so his performance evaluation rises while that of Carlos drops. Finally, **from 12:15 to 16:15**, Carlos has an AD first, however, he doesn't seize the last chance and Nicolas wins this set. As a result, Carlos's performance line of this period enjoys a dramatic rose and then a drop, while Nicolas's performance line has an exactly opposite trend.

5 Randomization Assessment of Momentum Impact

To overturn the skeptic that momentum doesn't have impact on the following games is to disclose the relationship between players' momentum and the swings in play or runs of success. As a result, conducting a comprehensive randomization assessment of momentum impact ought to meet the following requirements:

- The assessment can identify the differences between the random distribution of scores and real distribution of scores based on a quantitative and qualitative method.
- The assessment can vividly present whether accumulation of success during certain period can breed continuous success or not, just as what the "Snowball Effect" depicts.

5.1 The Assessment between Momentum and Swings in Play

To identify the differences of the distributions, especially those for the final match and all the matches in the 2023 Wimbledon Gentlemen's, the steps should be taken as follow:

Step 1: Apply Monte-Carlo Simulation to develop the random scores distribution. Monte-Carlo simulation is a statistical analysis method based on random sampling used to simulate the behavior of complex systems or processes.

To compare the differences and form a comprehensive understanding, for each match we analyze the scores frequency for each player separately. Take the final match of player 1 and player 2 for example. **First,** we process the data of each set's points and plot a histogram of the frequency of each point. **Then,** we utilize the Monte-Carlo simulation to form the forecast histogram of the frequency of each point if which is randomly scored. **Finally,** we get the 4 initial and 4 simulate frequency distribution histograms as follow:

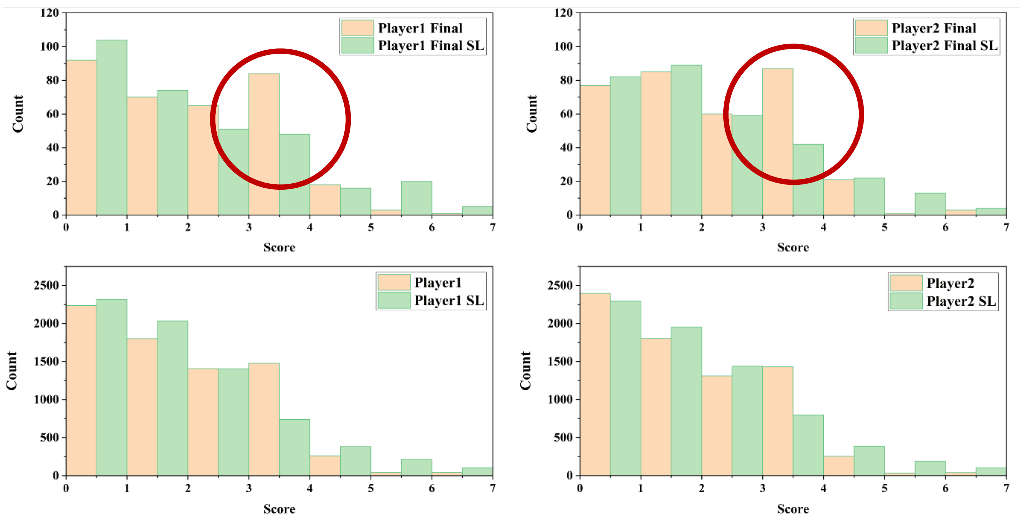

Figure 7: The Initial and Simulate Frequency Distribution Histograms of Scores

Step 2: Compare the distributions and describe some apparent differences among them. Among the 4 histograms, for example, in the 2023 Wimbledon men's tennis final, we can find that the real score distribution clearly does not match the simulated result distribution, such as the position marked in the picture.

Step 3: Utilize the Kolmogorov-Smirnov (KS) test to quantitatively measure the similarity between the distributions of corresponding initial and simulate scores. More specifically, we use the KS Statistic and P Value to quantify the similarities.

For P Value, it is a value that indicates the probability of observing the test results under the null hypothesis that the distributions of the two samples are the same. A small P value (typically⩽0.05) suggests that we can reject the null hypothesis, meaning the distributions are significantly different.

For KS Statistic, it is a value that quantifies the maximum distance between the empirical distribution functions of the two samples. A smaller value indicates more similar distributions. As a non-parametric test, the two-sample KS test, its core idea is to compare the cumulative distribution function (CDF) of two samples and calculate the maximum difference between them. Its advantage is that it has universal applicability independent of the type of data distribution and can be used for continuous or discrete data.

We use the actual score sample S_1 and simulated score sample S_2 of different column combinations to calculate KS statistics of different column combinations according to their cumulative distribution functions $F_{S_1}(x)$ and $F_{S_2}(x)$. The KS statistic D is defined as the greatest difference between these two distribution functions:

$$D = sup_x|F_{S_1}(x) - F_{S_2}(x)| \tag{19}$$

Where sup stands for "upper bound", that is, the maximum difference between two CDF among all possible x values.

After calculation, we have the results for the 4 groups as follow:

Table 8: The KS Statistic and P-value of the 4 Groups

	KS Statistic	P-value
p1_score_final vs p1_simulate_final	0.344	5.96×10^{-18}
p2_score_final vs p2_simulate_final	0.458	5.60×10^{-32}
p1_score vs p1_simulate	0.337	2.37×10^{-17}
p2_score vs p2_simualte	0.314	1.37×10^{-10}

Consequently, after conducting the procedures, we can safely refuse the null hypothesis that the distributions of the two samples are similar. In other words, the swings in play are not random, for the initial scores are not randomly distributed.

5.2 The Assessment between Momentum and Runs of Success

In tennis matches, one player's runs of success are invariably described as the consecutive points scored. On the website of ***US Open***, they describe the players' momentum as the consecutive points and draw a conclusion that "The more consecutive points won by a given player, the more momentum is in favor of that player." As a result, the relationship between momentum and runs of success can be made an analogy to the relationship between the players' performance and the point difference of the players.

Firstly, we sort out the dataset to apply the data related to the indicators system to the

PPEM and get the performance scores. **Then**, we draw the figure of player 1's performance scores during the whole match and the figure of the 2 players' points difference from the perspective of player 1. **Finally**, we compare the 2 figures and analyze the similar trends between them to get the conclusion. The figures we draw are presented as follow:

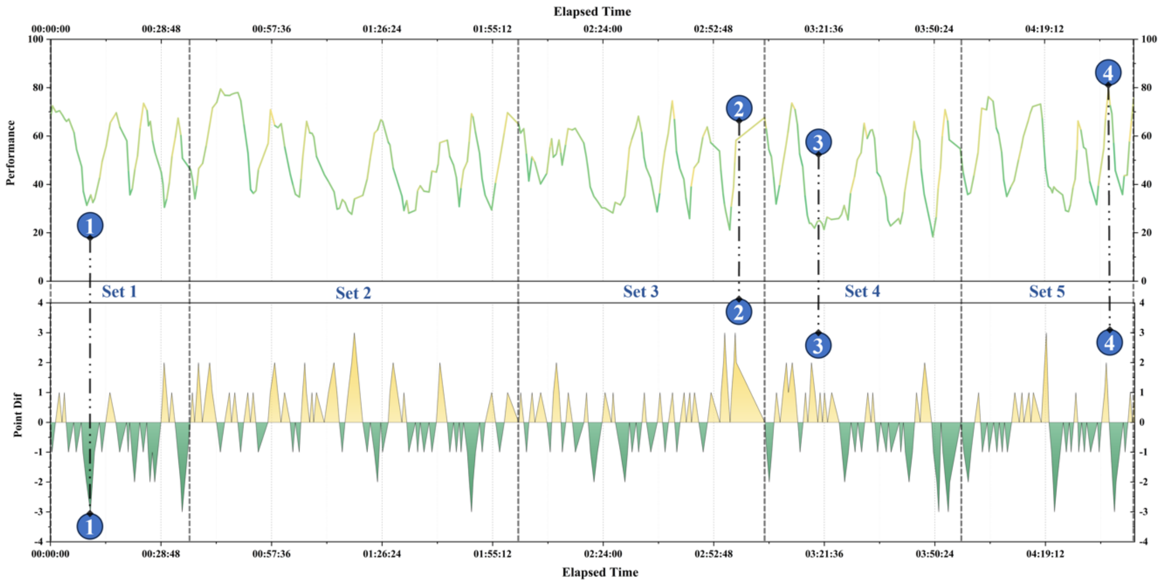

Figure 8: The Player 1' Performance Scores and Points Difference

From the 2 figures, we can notice that almost after every trend-change in player 1's performance comes with the trend-change in the 2 players' points difference. To be more specific, at point 1, the relatively worse performance of player 2 at the very beginning gives the player a negative momentum and this momentum can make the player's performance worse. At point 2, 3 and 4, we can find that positive momentum for player being reversed usually happens after the breaks in his own serve. The relatively better performance gives the player a positive momentum and this momentum can make the player's performance better.

Furthermore, we can notice that the trend-changes in players' performance anticipate the trend-changes in the players' point difference. This indirectly means that the changes in momentum can also be used to depict the changes in the future points scored, or to predict the swings in the match in another word.

Consequently, after conducting the procedures, we can conclude that the accumulation of success during certain period, usually reflected on the point difference of 2 players, can breed continuous success, which is also the impact of player's momentum. In other words, the runs of success are not random, and they are influenced by the momentum in some ways.

6 Momentum Prediction Model

Momentum, according to the previous analyses, has a great impact on the flow of play, which can give the player a greater advantage than the opponent to create some incredible swings in play. Fundamentally speaking, momentum is a word derived from the concept of momentum in physics, which stands for the tendency of an object to keep moving in the direction it is moving now. We have a formula to calculate momentum in physics as follows:

$$P = m \times v \tag{20}$$

Where P represents the momentum of the object, m represents the quality of the object and v represents the velocity of the object. In this case, m and v can be respectively implicated as the accumulation and the performance of the object.

As a result, we can make an analogy and form a preliminary inference that players' Momentum in sport matches can also be divided into 2 parts. One refers to the advantages accumulated before the beginning of the match, while the other refers to the relative performance during the match. According to the process of PPEM establishment to quantify players' performance, these can both be quantitatively represented by a series of indicators similarly. Based on this inference, we conduct more literature learning and problem analysis to establish the **Momentum Prediction Model (MPM).**

6.1 The Establishment of MPM

We introduce the Momentum (M) as the overall description of one player's general advantages over the other in the match. The main procedure to predict M is presented as follows:

The first step, conduct literature research to figure out the indicators that should be contained in the MPM.

The momentum comprises two distinct forms of momentum: psychological momentum and strategic momentum (Philippe Meier et al, 2020) [7]. Although both theories predict that success breeds success, the rationale behind these two theories is different (Gauriot and Page, 2018, 2019) [8,9].

For psychological momentum (PM), it suggests that a precipitating event triggers a performance increase due to changes in the perception of the agents during the event (e.g., Iso-Ahola & Mobily, 1980) [10]. After literature analysis, we conclude that the changes in PM of different players are mainly based on the perception of their own performance. As a result, PM can be evaluated by PPEM to great extent.

For strategic momentum (SM), it arises from the different relative positions of competing agents in a dynamic contest, which leads to asymmetric future expected prizes. As a result, we should choose indicators related to the "relative positions" of players. Furthermore, the indicators should evaluate the player before the beginning of the match. People have considered the indicator "Ranking" to be a decisive indicator However, according to the rule of ranking, it is mainly based on the points earned by each match and the closer to the final match, the more points the player will get. This rule overlooks the number of matches the player has player and the outcome of each match. Consequently, we creatively integrate the indicator "Win-loss Index", which is used to depict the ratio of the number of winning matches to the number of all matches the player having attended.

To better illustrate the PM and SM, we sort out the timeline as follows:

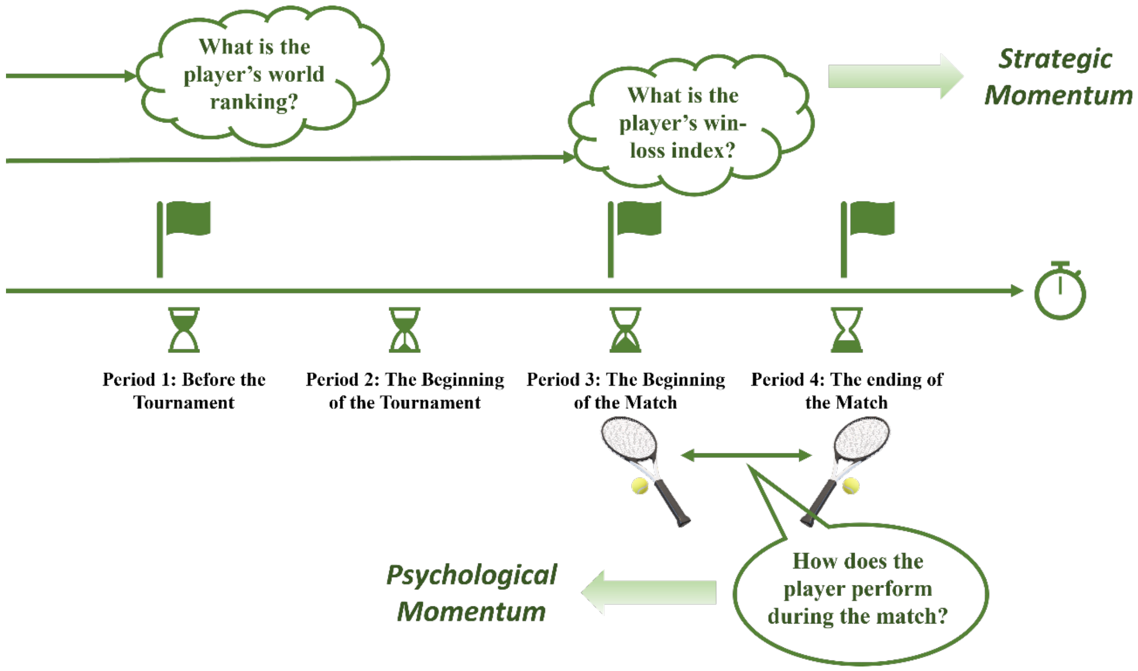

Figure 9: The Timeline to better illustrate SM and PM

The second step, optimize the PPEM by selecting more significant indicators to better evaluate the psychological momentum.

Select the indicators. Firstly, we compare the different Indicators' weights of PPEM again to eliminate less significant indicators. Secondly, we analyze the left indicators and the indicators selected by other literature to optimize our selection. Finally, we choose 7 indicators "Speed" "Server" "Unf Err" "Break Pt" "Leading Sets" "Leading Games" and "Winning Streak".

The third step, employ the method of machine learning to calculate the indicators' weights and establish the **Adjusted PPEM.** MPM is established by selecting the best method of 5 different methods of machine learning, Categorical Boosting (CATBoost), Light Gradient Boosting Machine (LGBM), Extreme Gradient Boosting (XGBoost), Random Forest, Logistic Regression (LR), by comparing the AUC value.

	Test AUC	Train AUC
LGBM	0.7074	0.7986
XGBoost	0.6960	0.8390
LR	0.7156	0.7082
Random Forest	0.6469	0.7223
CATBoost	0.7175	0.7170

Table 9: The Comparison of AUC

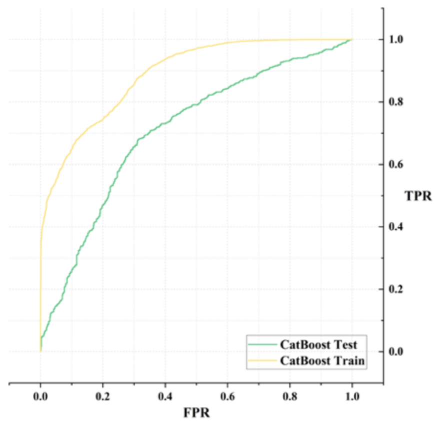

Figure 10: FPR-TPR

We choose the **Categorical Boosting (CATBoost)** to calculate the weights of indicators in MPM. Furthermore, considering that the tree-based model is particularly sensitive to the entropy of the input variables, and the entropy distribution of the original variable is quite different. Consequently, we impose a minimum order of magnitude random number based on the original variable to make the focus of the indicator flow to itself as much as possible. We get the indicators' weights as follow:

Table 10: The Weights of Indicators in the Adjusted PPEM

Indicators	Weights	Indicators	Weights
Speed	18.3841	Server	15.6998
Unf Err	14.2315	Break Pt	13.7250
Leading Sets	13.1307	Leading Games	12.9679
Winning Streak	11.8610		

Consequently, the Adjusted PPEM is presented as follows:

$$\begin{aligned} PM &= Adjusted\ PPEM \\ &= 18.3841 \times Speed + 15.6998 \times Server + 14.2315 \times Unf\ Err \\ &+ 13.7250 \times Break\ Pt + 13.1307 \times Leading\ Sets \\ &+ 12.9679 \times Leading\ Games + 11.8610 \times Winning\ Streak \end{aligned} \tag{21}$$

The fourth step, conduct a method similar with the third step to contain the indicators of SM into MPM and calculate the corresponding weights.

In order to get a more reasonable weight, we consult the website of Association of Tennis Professionals (ATP) to get the world rank of each player on the exact date of 2023.07.02, which is the closest to the beginning of the 2023 Wimbledon Gentlemen's and the number of win or loss matches to calculate the win-loss index.

Table 11: The Weights of Indicators in the Adjusted PPEM

Indicators	Weights	Indicators	Weights
Ranking	12.2514	Win-loss Index	13.7284

Finally, develop the MPM consisted of PM and EM. We have the final MPM as follows:

$$MPM = w_{PM} \times PM + w_{SM} \times SM = w_{PM} \times Adjusted\ PPEM + w_{SM} \times SM \tag{22}$$

From Momentum Prediction Model (MPM), we can apparently get that "Speed" is the most related indicator, for the largest weight 18.3841 it has.

6.2 The Advice for a Player Facing a New Match against a Different Player

The main purpose under this situation is to give a player advice to solve the **"3N" problem**, which refers to **"New opponent" "New match" and "New situation"**.

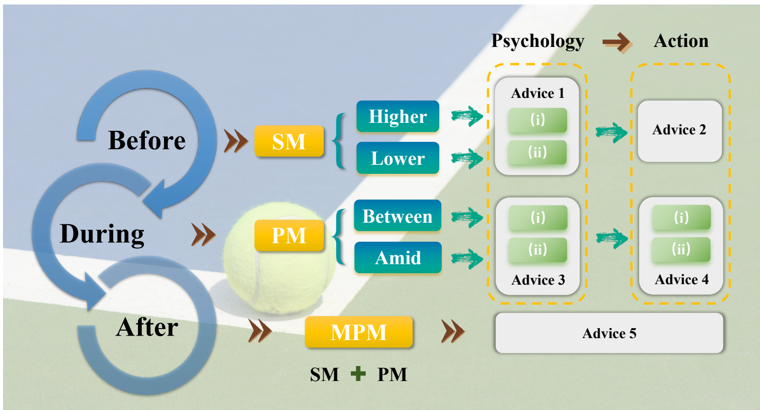

Figure 11: The Logic of Advice Proposed

Based on the MPM established above, the suggestions of players will be discussed from three aspects: pre-match, during match and post-match, just as what is presented as follows:

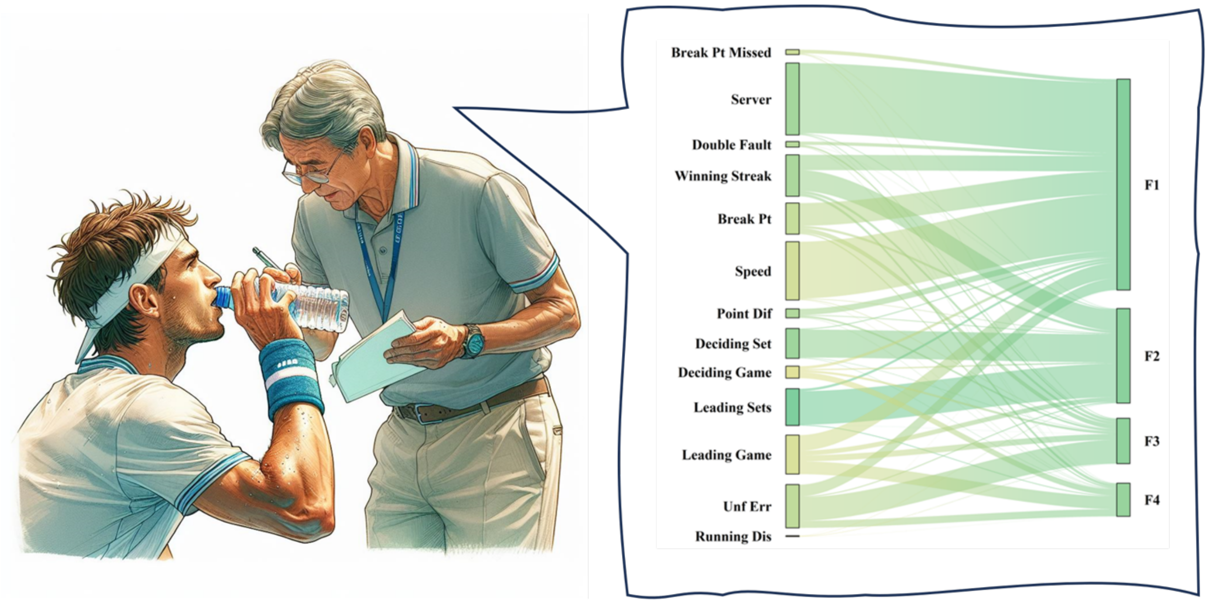

Figure 12: The visualization of Indicators' Weights

Advice 1: (i) For players with high rankings or excellent win-loss index, you should maintain a moderate amount of confidence, but not too arrogant, while maintaining sufficient respect for the strength of your opponent. This prevents erratic performance due to the outstanding performance of the opponent or the emergence of potential dark horse players. **(ii) For players with low rankings or low win-loss index**, self-confidence should be cultivated, and the strength of opponents should be recognized to develop appropriate game strategies. The coach should provide psychological counseling in these aspects to help the player maintain a stable mentality and performance in the game.

Advice 2: In the PM model, a player's speed, serve and receive direction are critical to momentum. Therefore, the coaching team should develop a comprehensive training strategy, focusing on the improvement of physical fitness and the improvement of serving and receiving skills. These trainings should complement match tactics to make up for shortfalls and optimize competitiveness, especially in building and maintaining momentum.

Advice 3: (i) Between each round, players are encouraged to take deep breaths and meditate during breaks to stay calm and focused, and to evaluate the performance of the previous round and learn from it. **(ii) Amid each round**, remind the player to stay positive, focus on each point, not affected by previous or future rounds, and focus on handling each serve and receive.

Advice 4: (i) After knowing the opponent, i.e., before each round, the coach should instruct the player to study the physical qualities and characteristics of the opponent in advance to better prepare for the match. **(ii) At the same time, during each round,** use reasonable pauses to adjust the game strategy according to changes in momentum. For example, increasing the offensive when the momentum is good and adopting a conservative strategy when the momentum is not good to improve the competitiveness of the game.

Advice 5: After each round and match, the coaching team should make use of data analysis tools to summarize and review the match in a timely manner. This allows for more accurate identification and use of momentum changes in a match, and more targeted strategies for athletes to prepare for subsequent rounds and matches.

7 Generalization Ability Test for Momentum Prediction Model

According to the basic rules of generalization, our Momentum prediction model should be applicable to different kinds of sport matches, irrespective of players' gender, types of balls, court surfaces etc. In order to conduct the generalization ability test for MPM, we apply the data of 2 matches, one of the matches between Iga Swiatek and Rebecca Peterson in 2023 US Women's Open and the table tennis final between MA Long and FAN Zhendong in 2021 Tokyo Olympics.

7.1 Generalization Ability Test for MPM in Tennis of Women's Single

After we apply the data of the women's tennis match into the MPM, we discover that the trends of players' momentum and the trends of point difference are nearly the same, which represents a great generalization ability of the MPM in the tennis women's single on the clay court.

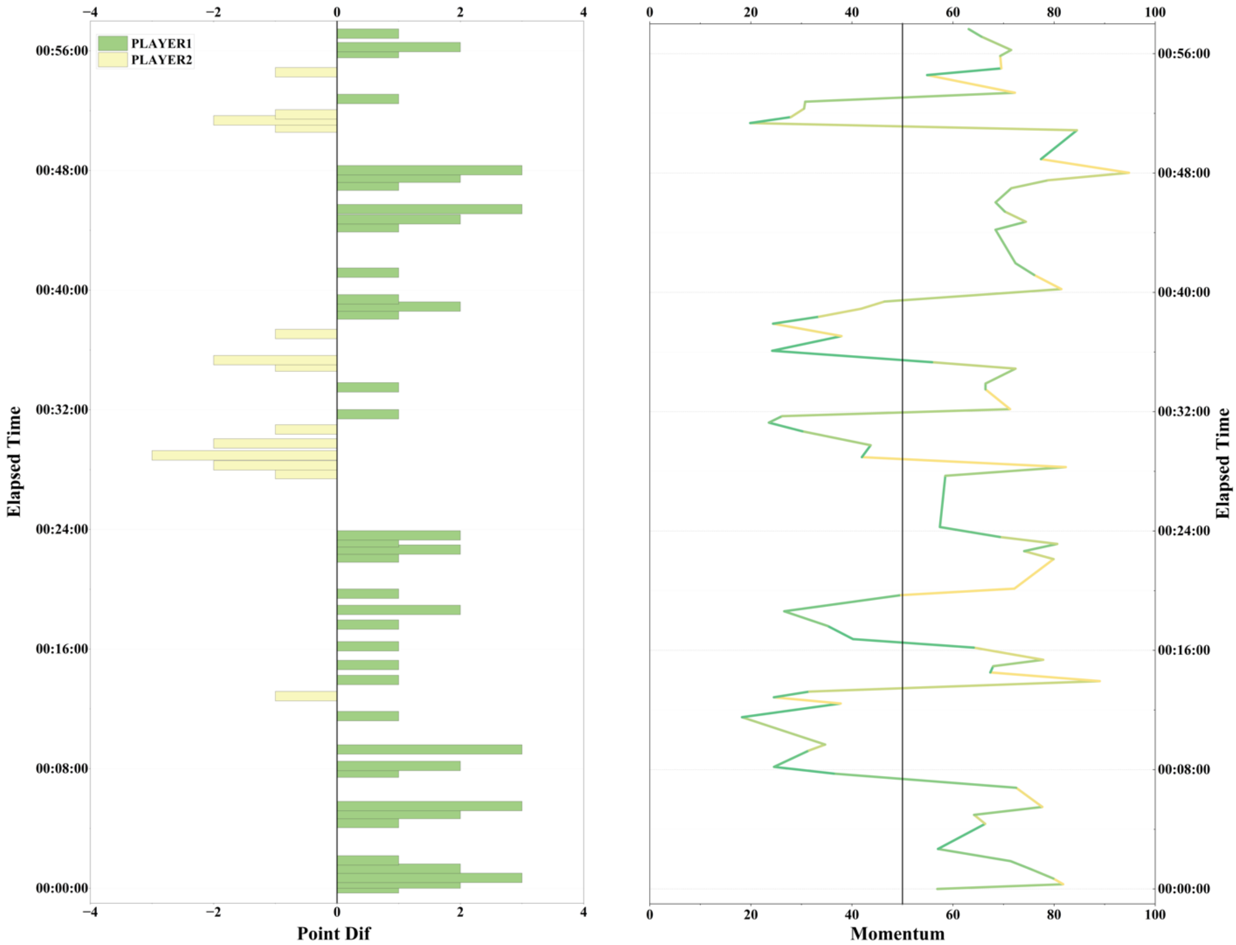

Figure 13: Point Difference and Momentum Prediction in Tennis Women's Single

However, at the beginning of the match, there is some opposite trends of point difference and momentum, because there are some different features in tennis match compared women to men. **Hence**, in the future, we will combine more indicators more related to women, such as the gender and adjust the weights of different indicators.

7.2 Generalization Ability Test for MPM in Table Tennis of Men's Single

After we apply the data of the Olympics men's table tennis match into the MPM, we discover that the trends of players' momentum and the trends of point difference are nearly the same, which represents a great generalization ability of the MPM in the tennis women's single.

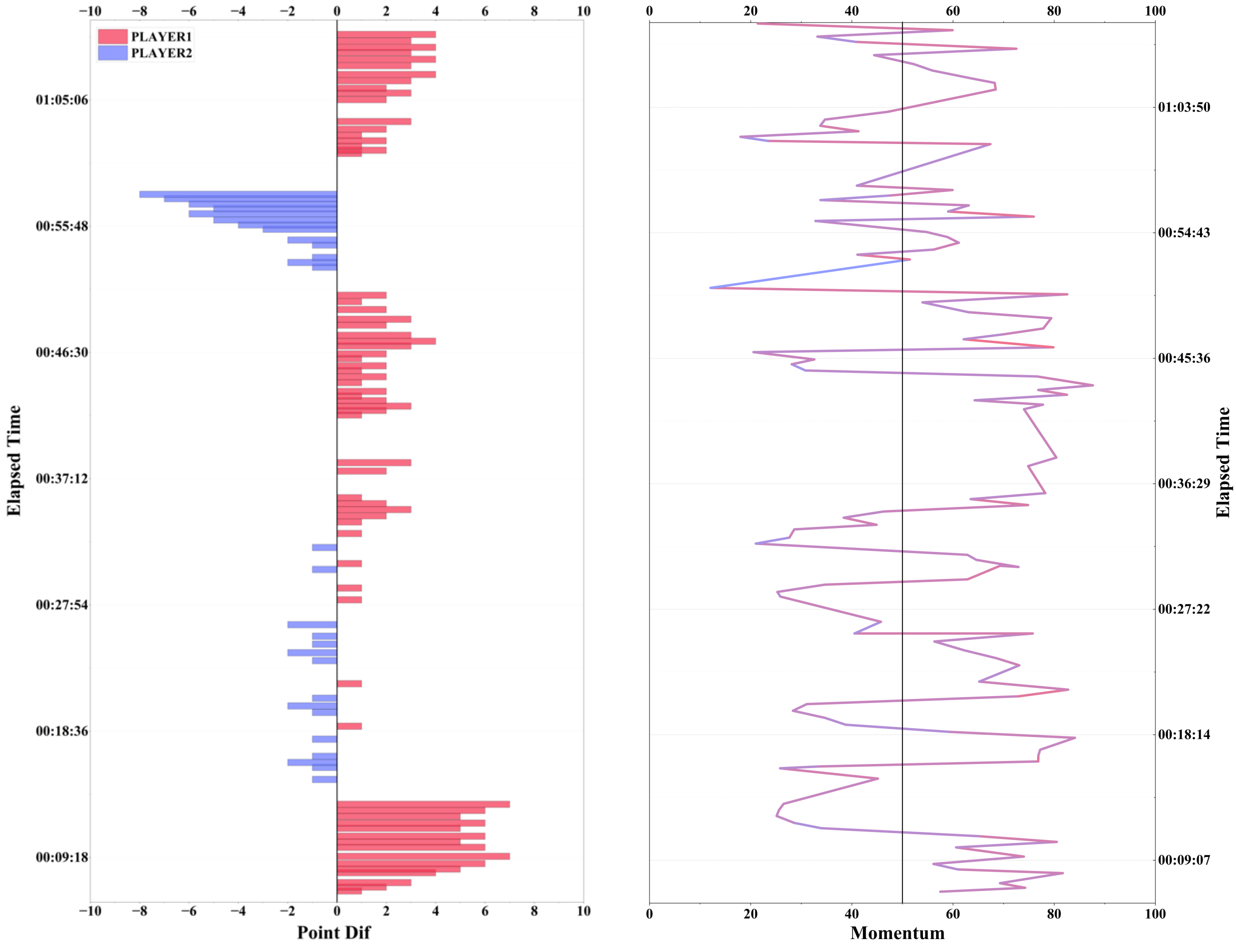

Figure 14: Point Difference and Momentum Prediction in Table Tennis Men's Single

However, at the middle of the match, there is some opposite trends of point difference and momentum, because there are some different features in tennis match when compared to table tennis. **Hence**, in the future, we will combine more indicators more related to table tennis, such as the rotations of the ball, because for a small ball, a small change in wrist action can greatly change the direction of the ball and where it lands.

Consequently, from the tests we conduct, we can make an inference that our MPM is greatly generalizable to most sport matches. More poor predictions can be eradicated by taking more indicators into consideration and adjusting the weights of indicators under different circumstances.

8 Sensitivity Analysis

We adjust the weight factors of PPEM key factors in Model 1 and find that the model can produce similar results under multiple parameter changes, which indicates that the model is relatively stable and is not susceptible to interference from small parameter changes. At the same time, consistent decision results enhance the reliability of the model, because different parameter values lead to similar results, which means that the output of the model is reliable.

The weight of PM index in Model 2 is trained by using the data of men's singles match in 2023 Wimbledon. Will the weight of each index change significantly if only one indicator is changed? The training set is changed to the women's singles match in 2023. Only Server and Speed indicators have a larger change (-13.60%, -9.73%), which can be considered as a greater explosion of male players on the serve, so the two indicators have a greater impact on the momentum of male players than female players. It can be argued that the model also needs to supplement the "gender" factor.

In order to better depict the changes is indicators' weights in sensitivity analysis, we draw the radar figures as follow:

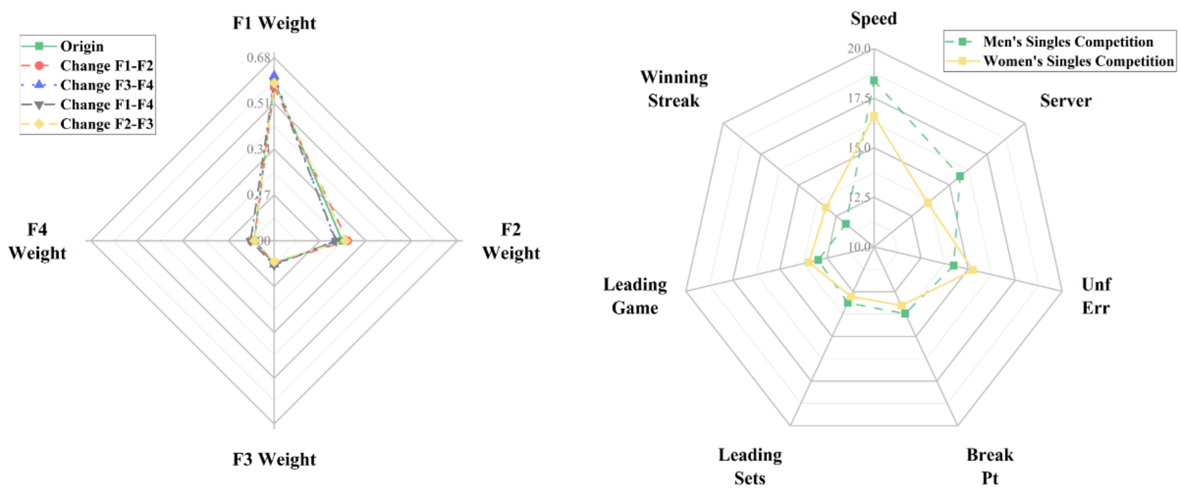

Figure 15: The Radar Chart of Value Comparison after Parameter Changing

9 Model Evaluation

9.1 Strengths

- **More comprehensive indicators are considered in models.** The PPEM considers most of the significant and principal indicators influencing the players' performance.
- **More advantageous methods are employed in establishing models.** The combined use of PCA and AHP helps achieve a dimensionality reduction and weights calculation by transforming large quantities of indicators into some principal components. Furthermore, the outliers which are more sensitive to the model are also treated.
- **Broader perspectives are utilized in optimizing models.** The MPM doesn't limit momentum into just psychological momentum reflected during the match i.e., PM and further consider the strategic momentum accumulated before the match i.e., SM.

9.2 Weaknesses

- **Extra data collecting is time-consuming and difficult to get the official ones.** The ranking and win-loss index considered in SM of MPM should call for a detailed search in different websites, which is apt to make mistakes and accordingly influence the accuracy of the models.

- **There is some room for improvement in selection of indicators and determination of indicators' weights.** To further improve the generalization ability of our model, we should take more indicators into consideration under different circumstances of different matches, such as gender and type of the ball.

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Resources: US OPNE official website, WTT official website, ATP official website

Memorandum

To: Coaches who may concern
From: Team # 2415017
Date: February 5, 2024
Subject: Advice for coaching from the perspective of momentum

Dear coaches,

We are an experienced sports analytics team from MCM, and it's a great honor to present our insights on the role of momentum and strategic recommendations for coaching with you.

Our analysis has found that despite increasingly fierce competition and unpredictable outcomes, the incredible influence of "momentum" is certain.

Firstly, through literature review and data pre-processing, we have established a Player Performance Evaluation Model (PPEM) using Principal Component Analysis and Analytic Hierarchy Process, which vividly displays players' performance and strengths.

Secondly, we utilized Monte-Carlo simulation and the Kolmogorov-Smirnov test to evaluate fluctuations within the match and analyze their relationship with score differentials, ultimately discovering that the continuity of success is not random and is indeed affected by momentum.

Thirdly, combining Strategic Momentum (SM) and Psychological Momentum (PM), we developed a Momentum Prediction Model (MPM). By refining PPEM and eliminating less significant indicators, we depicted PM, while SM was approached by combining win-loss indexes and rankings. Based on the MPM, we provided players with suggestions to tackle the "3N" issues (New opponent, New match, New situation).

Lastly, we tested the generalization ability of MPM using data from women's singles tennis and men's singles table tennis matches, proposing directions for future model optimization.

Based on the mentioned analysis, our final recommendations are as follow:

1. Regardless of ranking or win-loss index, you need to help players maintain a moderate level of confidence and respect for their opponents' abilities to sustain a stable state and performance in the game.

2. In the PM model, the speed of play, server, and receiving directions are critical. The coaching team should develop a comprehensive training strategy to enhance physical fitness and skills, especially in establishing and maintaining momentum.

3. Players are encouraged to take deep breaths and meditate during breaks to stay calm and focused on each point, unaffected by previous or subsequent rounds.

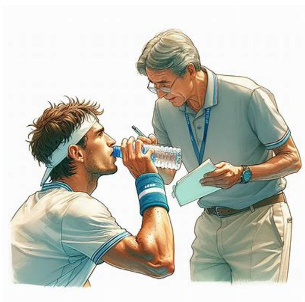
4. Coaches should guide players to study the physical qualities and characteristics of opponents in advance and adjust strategies timely to respond to momentum swings.

5. After each round and match, the coaching team should use data analysis tools to summarize promptly, providing targeted strategies for subsequent games.

We hope our suggestions are helpful and wish you success in the competitions! Please let us know if you have any questions.

Kinds and regards,
Team # 2415017

Report on Use of AI

1. OpenAI ChatGPT (Feb 2, 2024 version, ChatGPT4,)
   Query1: Please help me generate a series of color schemes related to table tennis.
   Output: Here are five color palettes related to table tennis, each comprising four colors:
   Palette 1: White (#FFFFFF), Table Green (#00A170), Wood Floor (#8B4513), Boundary Blue (#3C3CFF)
   Palette 2: Table Green (#00A170), Net Gray (#A1A1A1), Boundary Blue (#3C3CFF), White (#FFFFFF)
   Palette 3: Racket Red (#C8102E), Boundary Blue (#3C3CFF), White (#FFFFFF), Table Green (#00A170)
   Palette 4: Ball Orange (#FFA500), White (#FFFFFF), Wood Floor (#8B4513), Racket Black (#000000)
   Palette 5: Net Gray (#A1A1A1), Ball Orange (#FFA500), Wood Floor (#8B4513), Racket Black (#000000)
   These palettes combine the iconic colors of table tennis equipment and environment, suitable for designing sports apparel, equipment, or promotional materials related to the sport.
2. OpenAI ChatGPT (Feb 2, 2024 version, ChatGPT4,)
   Query1: Please draw a scene where a tennis coach guides athletes during a game break, with a pure white background and a realistic style. The color scheme should be as close to the two colors (# 5EC286) and (# FAE282) related to tennis matches as possible. Additional condition: 1. As the time point is during the game break, the athlete should be slightly sweating and listening to the coach's guidance while drinking water. You should highlight the guidance screen, and the coach should hold a notebook.
   Output:   
3. OpenAI ChatGPT (Feb 2, 2024 version, ChatGPT4,)
   Query1: May I ask, in tennis matches, besides the athlete's own factors, what other factors may affect the athlete's performance and momentum?
   Output: In tennis matches, besides the athlete's own factors, there are also some other factors that may affect the athlete's performance and momentum, such as:

   Physical factors: Tennis matches usually last for several hours and require high physical fitness from athletes. If an athlete's physical fitness is insufficient, or they experience fatigue, injury, etc. during the competition, it will affect their skills, speed, strength, and endurance, thereby reducing their performance and momentum.

Psychological factors: In tennis matches, athletes need to face various pressures and challenges, such as opponent's strength, changes in scores, audience reactions, etc. If athletes have weak psychological qualities or experience emotions such as tension, anxiety, and fear during the competition, it will affect their judgment, decision-making, and execution, thereby reducing their performance and momentum.
Technical and tactical factors: In tennis matches, athletes need to develop appropriate technical and tactical plans based on their own characteristics and the characteristics of their opponents, such as serving, receiving, baseline, surfing, defense, etc. If the skills and tactics of athletes are unreasonable or cannot be flexibly adjusted during the competition, it will affect their strengths and weaknesses, thereby reducing their performance and momentum.
External environmental factors: In tennis matches, athletes need to adapt to different types of events, surface conditions of the court, climate conditions, and the judgments of referees and other external environmental factors. If athletes are not familiar with these factors or cannot adapt in a timely manner during the competition, it will affect their feelings, rhythm, and confidence, thereby reducing their performance and momentum.

The above are some factors that I have summarized for you based on online literature that may affect the performance and momentum of athletes in tennis matches, hoping to be helpful to you.